



Habib University
shaping futures

Instructions

1. The exam carries **50 marks** in total.
2. This is a closed books/notes exam. Any notes, books, bags should be kept at the front of the classroom before the start of the test.
3. Use of cellphones, laptops or any other communicating device is prohibited. Your cellphones should be switched off and submitted to the invigilator, or kept at the front of the class room before the start of the test.
4. You are to attempt all questions on the answer sheet provided.
5. Your solutions should be comprehensible. Explain what you are doing, and if you use any results that you have studied in this course or Calculus 1 and 2, mention them explicitly.
6. Use of calculators is allowed for this exam.
7. Any form of communication/sharing among your peers is not allowed during the exam.
8. If you want to communicate with the invigilator, raise your hand.
9. You are not allowed to leave your desk without permission of the invigilator during the exam.
10. Failure to abide by the instructions above will lead to immediate cancellation of the exam.

Student Name: _____

Instructor's Name: _____

Section: _____

1. (6 points) Solve the following differential equation:

$$(x + y) \sin y \, dx + (x \sin y + \cos y) \, dy = 0.$$

Solution: Given the differential equation:

$$(x + y) \sin y \, dx + (x \sin y + \cos y) \, dy = 0,$$

Step 1: Identify $M(x, y)$ and $N(x, y)$

We rewrite the equation in the standard form:

$$M(x, y) \, dx + N(x, y) \, dy = 0,$$

where:

$$M(x, y) = (x + y) \sin y, \quad N(x, y) = x \sin y + \cos y.$$

Step 2: Check for Exactness

To check for exactness, compute:

$$\frac{\partial M}{\partial y} \quad \text{and} \quad \frac{\partial N}{\partial x}.$$

First, calculate $\frac{\partial M}{\partial y}$:

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y} ((x + y) \sin y) = (x + y) \cos y + \sin y.$$

Next, calculate $\frac{\partial N}{\partial x}$:

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} (x \sin y + \cos y) = \sin y.$$

Since:

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x},$$

the equation is not exact.

Step 3: Introduce an Integrating Factor

Consider

$$\begin{aligned} R(y) &= \frac{N_x - M_y}{M} \\ &= \frac{\sin(y) - x \cos(y) - y \cos(y) - \sin(y)}{(x + y) \sin(y)} \\ &= -\frac{\cos(y)}{\sin(y)} \end{aligned}$$

The integrating factor is

$$\begin{aligned} \mu(y) &= \exp \left(\int -\frac{\cos(y)}{\sin(y)} dy \right) \\ &= \exp(-\ln(\sin(y))) \\ &= \frac{1}{\sin(y)} \end{aligned}$$

We test the integrating factor $\mu(y) = \frac{1}{\sin y}$. Multiplying through by $\mu(y)$, the equation becomes:

$$\frac{1}{\sin y} ((x + y) \sin y \, dx + (x \sin y + \cos y) \, dy) = 0,$$

which simplifies to:

$$(x + y) \, dx + (x + \cot y) \, dy = 0.$$

Now:

$$M(x, y) = x + y, \quad N(x, y) = x + \cot y.$$

Step 4: Check for Exactness Again

Compute $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$:

$$\frac{\partial M}{\partial y} = 1, \quad \frac{\partial N}{\partial x} = 1.$$

Since:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

the equation is now exact.

Step 5: Solve the Exact Equation

For an exact equation, there exists a potential function $\Phi(x, y)$ such that:

$$\frac{\partial \Phi}{\partial x} = M(x, y), \quad \frac{\partial \Phi}{\partial y} = N(x, y).$$

From $\frac{\partial \Phi}{\partial x} = x + y$, integrate with respect to x :

$$\Phi(x, y) = \frac{x^2}{2} + xy + h(y),$$

where $h(y)$ is an arbitrary function of y .

From $\frac{\partial \Phi}{\partial y} = x + \cot y$, substitute $\Phi(x, y)$:

$$\frac{\partial}{\partial y} \left(\frac{x^2}{2} + xy + h(y) \right) = x + \cot y.$$

Simplify:

$$x + h'(y) = x + \cot y.$$

Thus:

$$h'(y) = \cot y.$$

Integrate $h'(y)$ with respect to y :

$$h(y) = \ln |\sin y| + C,$$

where C is a constant.

The potential function is:

$$\Phi(x, y) = \frac{x^2}{2} + xy + \ln |\sin y| + C.$$

Step 6: Solution

The solution to the differential equation is given implicitly by:

$$\Phi(x, y) = \text{constant},$$

or:

$$\frac{x^2}{2} + xy + \ln |\sin y| = C.$$

2. (8 points) Solve the initial value problem:

$$y'' + y = 8 \cos 2x - 4 \sin x, \quad (1)$$

with the initial conditions:

$$y\left(\frac{\pi}{2}\right) = -1, \quad y'\left(\frac{\pi}{2}\right) = 0.$$

- 2.1 (2 Points) Find the general solution, y_c , of the associated homogeneous equation $y'' + y = 0$.
- 2.2 (2 Points) Suggest a guess function y_p for the non-homogeneous part $r(x) = 8 \cos 2x - 4 \sin x$. Justify your answer.
- 2.3 (3 Points) Solve for y_p using the Method of Undetermined Coefficients.
- 2.4 (0.5 Points) Write down the general solution of the second-order non-homogeneous equation (1).
- 2.5 (0.5 Points) Solve the initial value problem.

Solution:

We solve the differential equation:

$$y'' + y = 8 \cos(2x) - 4 \sin(x),$$

with the initial conditions:

$$y\left(\frac{\pi}{2}\right) = -1, \quad y'\left(\frac{\pi}{2}\right) = 0.$$

2.1: General Solution of the Homogeneous Equation

The homogeneous equation is:

$$y'' + y = 0.$$

The characteristic equation is:

$$r^2 + 1 = 0 \quad \implies \quad r = \pm i.$$

Thus, the general solution of the homogeneous equation is:

$$y_c(x) = C_1 \cos(x) + C_2 \sin(x),$$

where C_1 and C_2 are constants.

2.2: Particular Solution $y_p(x)$

The non-homogeneous term is:

$$r(x) = 8 \cos(2x) - 4 \sin(x).$$

Since the $\sin(x)$ term overlaps with the solution to the homogeneous equation, the particular solution must account for this. The appropriate guess is:

$$y_p(x) = A \cos(2x) + B \sin(2x) + x(C \cos(x) + D \sin(x)).$$

2.3: Finding Particular Solution

Step 1: Derivatives of $y_p(x)$

The first derivative of $y_p(x)$ is:

$$y_p'(x) = -2A \sin(2x) + 2B \cos(2x) + C(\cos(x) - x \sin(x)) + D(\sin(x) + x \cos(x)).$$

The second derivative of $y_p(x)$ is:

$$y_p''(x) = -4A \cos(2x) - 4B \sin(2x) + C(-2 \sin(x) - x \cos(x)) + D(2 \cos(x) - x \sin(x)).$$

Step 2: Substituting into the Differential Equation

Substitute $y_p(x)$, $y_p'(x)$, and $y_p''(x)$ into $y'' + y = 8 \cos(2x) - 4 \sin(x)$. The left-hand side becomes:

$$\begin{aligned} y_p''(x) + y_p(x) &= (-4A \cos(2x) - 4B \sin(2x)) + (A \cos(2x) + B \sin(2x)) \\ &+ x(-C \cos(x) + D \sin(x)) + x(C \cos(x) + D \sin(x)) + C(-2 \sin(x)) + D(2 \cos(x)). \end{aligned}$$

Combine terms:

$$y_p''(x) + y_p(x) = (-4A + A) \cos(2x) + (-4B + B) \sin(2x) + (C \cdot -2) \sin(x) + (D \cdot 2) \cos(x).$$

Simplify:

$$y_p''(x) + y_p(x) = -3A \cos(2x) - 3B \sin(2x) - 2C \sin(x) + 2D \cos(x).$$

Equate the coefficients on the left-hand side to the non-homogeneous term $8 \cos(2x) - 4 \sin(x)$:

$$-3A \cos(2x) - 3B \sin(2x) - 2C \sin(x) + 2D \cos(x) = 8 \cos(2x) - 4 \sin(x).$$

Step 3: Solve for Coefficients

From the coefficients of $\cos(2x)$:

$$-3A = 8 \quad \implies \quad A = -\frac{8}{3}.$$

From the coefficients of $\sin(2x)$:

$$-3B = 0 \quad \implies \quad B = 0.$$

From the coefficients of $\sin(x)$:

$$-2C = -4 \quad \implies \quad C = 2.$$

From the coefficients of $\cos(x)$:

$$2D = 0 \quad \implies \quad D = 0.$$

Step 4: Particular Solution

Substitute the coefficients into the guess for $y_p(x)$:

$$y_p(x) = A \cos(2x) + B \sin(2x) + x(C \cos(x) + D \sin(x)).$$

With $A = -\frac{8}{3}$, $B = 0$, $C = 2$, and $D = 0$, this becomes:

$$y_p(x) = -\frac{8}{3} \cos(2x) + 2x \cos(x).$$

2.4: General Solution of the Differential Equation

The general solution is the sum of the homogeneous and particular solutions:

$$y(x) = y_c(x) + y_p(x),$$

where:

$$y(x) = C_1 \cos(x) + C_2 \sin(x) + 2x \cos(x) - \frac{8}{3} \cos(2x).$$

2.5: Apply Initial Conditions

We are given:

$$y\left(\frac{\pi}{2}\right) = -1, \quad y'\left(\frac{\pi}{2}\right) = 0.$$

Step 1: Apply $y\left(\frac{\pi}{2}\right) = -1$

Substitute $x = \frac{\pi}{2}$ into $y(x)$:

$$y\left(\frac{\pi}{2}\right) = C_1 \cos\left(\frac{\pi}{2}\right) + C_2 \sin\left(\frac{\pi}{2}\right) + 2 \cdot \frac{\pi}{2} \cos\left(\frac{\pi}{2}\right) - \frac{8}{3} \cos\left(2 \cdot \frac{\pi}{2}\right).$$

Using $\cos\left(\frac{\pi}{2}\right) = 0$, $\sin\left(\frac{\pi}{2}\right) = 1$, and $\cos(\pi) = -1$, this simplifies to:

$$-1 = 0 + C_2 + 0 - \frac{8}{3}(-1).$$

$$-1 = C_2 + \frac{8}{3}.$$

$$C_2 = -1 - \frac{8}{3} = -\frac{11}{3}.$$

Step 2: Apply $y'\left(\frac{\pi}{2}\right) = 0$

The derivative of $y(x)$ is:

$$y'(x) = -C_1 \sin(x) + C_2 \cos(x) + 2(\cos(x) - x \sin(x)) + \frac{16}{3} \sin(2x).$$

Substitute $x = \frac{\pi}{2}$:

$$y'\left(\frac{\pi}{2}\right) = -C_1 \sin\left(\frac{\pi}{2}\right) + C_2 \cos\left(\frac{\pi}{2}\right) + 2\left(\cos\left(\frac{\pi}{2}\right) - \frac{\pi}{2} \sin\left(\frac{\pi}{2}\right)\right) + \frac{16}{3} \sin(\pi).$$

Using $\sin\left(\frac{\pi}{2}\right) = 1$, $\cos\left(\frac{\pi}{2}\right) = 0$, and $\sin(\pi) = 0$, this simplifies to:

$$0 = -C_1 + 0 + 2\left(0 - \frac{\pi}{2}\right) + 0.$$

$$0 = -C_1 - \pi.$$

$$C_1 = -\pi.$$

2.6: Final Solution

The final solution to the initial value problem is:

$$y(x) = -\pi \cos(x) - \frac{11}{3} \sin(x) + 2x \cos(x) - \frac{8}{3} \cos(2x).$$

3. (10 points) Answer the questions below for the following initial value problem

$$y'' + y' - 6y = 0, \quad y(0) = 10, \quad y'(0) = 0$$

- 3.1 (1 Point) Write down the guess for $y(x)$ as a power series. Also determine $y'(x)$ and $y''(x)$ as power series.
- 3.2 (1 Point) Use the initial conditions to find the first two coefficients of the power series for $y(x)$.
- 3.3 (4 Points) Substitute your guess for $y(x)$, $y'(x)$, and $y''(x)$ into the ODE and simplify.
- 3.4 (2 Points) Find the recurrence relation for the coefficients, i.e., write a_{n+2} in terms of a_{n+1} and a_n , $n = 0, 1, 2, \dots$
- 3.5 (2 Points) Write down the general solution $y(x)$ as a power series with at least five terms.

Solution: To solve the differential equation

$$y'' + y' - 6y = 0$$

with the initial conditions $y(0) = 10$ and $y'(0) = 0$ using the power series method, we proceed as follows.

Step 1: Assume a power series solution

Assume the solution is in the form of a power series:

$$y(x) = \sum_{n=0}^{\infty} a_n x^n.$$

The first and second derivatives of $y(x)$ are:

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

Step 2: Substitute into the differential equation

Substitute $y(x)$, $y'(x)$, and $y''(x)$ into the differential equation $y'' + y' - 6y = 0$:

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=1}^{\infty} n a_n x^{n-1} - 6 \sum_{n=0}^{\infty} a_n x^n = 0.$$

Reindex the first two sums:

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n,$$

$$\sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1)a_{n+1} x^n.$$

Now the equation becomes:

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)a_{n+1} - 6a_n] x^n = 0.$$

For this equation to hold for all x , each coefficient of x^n must be zero. Therefore, we obtain the recurrence relation:

$$(n+2)(n+1)a_{n+2} + (n+1)a_{n+1} - 6a_n = 0.$$

Step 3: Use the initial conditions

From the initial conditions $y(0) = 10$ and $y'(0) = 0$, we can determine the first two coefficients:

$$y(0) = 10 \quad \Rightarrow \quad a_0 = 10,$$

$$y'(0) = 0 \quad \Rightarrow \quad a_1 = 0.$$

Step 4: Solve the recurrence relation

The recurrence relation is:

$$(n+2)(n+1)a_{n+2} = 6a_n - (n+1)a_{n+1}.$$

Now, compute the first few terms:

- For $n = 0$:

$$(0 + 2)(0 + 1)a_2 = 6a_0 - (0 + 1)a_1$$

$$\Rightarrow 2 \cdot 1 \cdot a_2 = 6 \cdot 10 - 1 \cdot 0 \Rightarrow 2a_2 = 60 \Rightarrow a_2 = 30.$$

- For $n = 1$:

$$(1 + 2)(1 + 1)a_3 = 6a_1 - (1 + 1)a_2$$

$$\Rightarrow 3 \cdot 2 \cdot a_3 = 6 \cdot 0 - 2 \cdot 30 \Rightarrow 6a_3 = -60 \Rightarrow a_3 = -10.$$

- For $n = 2$:

$$(2 + 2)(2 + 1)a_4 = 6a_2 - (2 + 1)a_3$$

$$\Rightarrow 4 \cdot 3 \cdot a_4 = 6 \cdot 30 - 3 \cdot (-10) \Rightarrow 12a_4 = 180 + 30 \Rightarrow a_4 = 35/2.$$

- For $n = 3$:

$$(3 + 2)(3 + 1)a_5 = 6a_3 - (3 + 1)a_4$$

$$\Rightarrow 5 \cdot 4 \cdot a_5 = 6 \cdot (-10) - 4 \cdot 17.5 \Rightarrow 20a_5 = -60 - 70 \Rightarrow a_5 = -13/2.$$

Step 5: Write the power series solution

Now that we have the first few coefficients, the power series solution is:

$$y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \dots$$

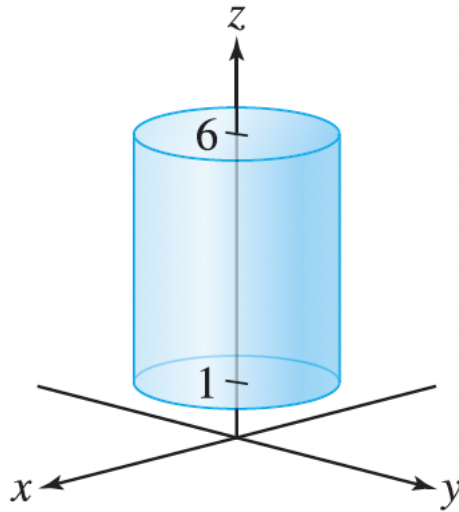
Substitute the values of the coefficients:

$$y(x) = 10 + 0 \cdot x + 30x^2 - 10x^3 + 17.5x^4 - 6.5x^5 + \dots$$

4. (12 points) Let $\mathcal{S}$ be the surface of the cylinder (not including the top and bottom) of radius 2 for $1 \leq z \leq 6$, oriented with outward-pointing normal.

4.1 Indicate with an arrow the positive orientation of ∂S (the top and bottom circles).

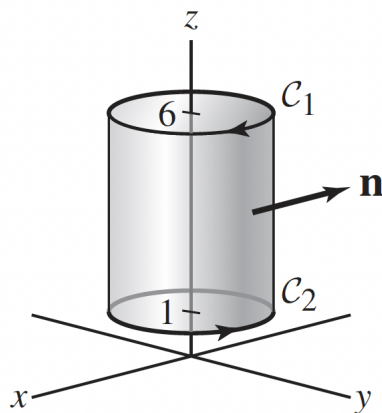
4.2 Verify the Stokes' Theorem for S and $F = yz^2\mathbf{i}$



Solution:

1. Induced Orientation of the Boundary

The induced orientation is defined so that, as the normal vector travels along the boundary curve, the surface lies to its left. Therefore, the boundary circles on the top and bottom of the cylinder have opposite orientations, as shown in the figure. Let C_1 denote the top circle ($z = 6$) and C_2 denote the bottom circle ($z = 1$).



2. Verification of Stokes' Theorem for S and $\mathbf{F} = yz^2\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$

Step 1: Compute the Integral Around the Boundary Circles

We use the following parametrizations:

$$C_1 : \gamma_1(t) = 2 \cos t \mathbf{i} + 2 \sin t \mathbf{j} + 6 \mathbf{k}, \quad t \in [2\pi, 0],$$

$$C_2 : \gamma_2(t) = 2 \cos t \mathbf{i} + 2 \sin t \mathbf{j} + 1 \mathbf{k}, \quad t \in [0, 2\pi].$$

For C_1 :

$$\mathbf{F}(\gamma_1(t)) = yz^2 \mathbf{i} + 0 \mathbf{j} + 0 \mathbf{k} = 72 \sin t \mathbf{i},$$

$$\frac{d\gamma_1(t)}{dt} = -2 \sin t \mathbf{i} + 2 \cos t \mathbf{j} + 0 \mathbf{k}.$$

The dot product is:

$$\mathbf{F}(\gamma_1(t)) \cdot \frac{d\gamma_1(t)}{dt} = (72 \sin t \mathbf{i}) \cdot (-2 \sin t \mathbf{i} + 2 \cos t \mathbf{j} + 0 \mathbf{k}) = -144 \sin^2 t.$$

For C_2 :

$$\mathbf{F}(\gamma_2(t)) = yz^2 \mathbf{i} + 0 \mathbf{j} + 0 \mathbf{k} = 2 \sin t \mathbf{i},$$

$$\frac{d\gamma_2(t)}{dt} = -2 \sin t \mathbf{i} + 2 \cos t \mathbf{j} + 0 \mathbf{k}.$$

The dot product is:

$$\mathbf{F}(\gamma_2(t)) \cdot \frac{d\gamma_2(t)}{dt} = (2 \sin t \mathbf{i}) \cdot (-2 \sin t \mathbf{i} + 2 \cos t \mathbf{j} + 0 \mathbf{k}) = -4 \sin^2 t.$$

The line integral around ∂S is:

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r}.$$

Substituting the results:

$$\begin{aligned} \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} &= \int_{2\pi}^0 (-144 \sin^2 t) dt + \int_0^{2\pi} (-4 \sin^2 t) dt \\ &= \int_0^{2\pi} 140 \sin^2(t) dt \end{aligned}$$

Using $\sin^2 t = \frac{1 - \cos(2t)}{2}$, the integrals become:

$$\int_0^{2\pi} 140 \sin^2 t dt = 140 \int_0^{2\pi} \frac{1 - \cos(2t)}{2} dt = 140 \cdot \pi,$$

Thus:

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = 140\pi$$

Step 2: Compute the Curl of $\mathbf{F}$

The curl of $\mathbf{F}$ is:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz^2 & 0 & 0 \end{vmatrix} = 0\mathbf{i} + 2yz\mathbf{j} - z^2\mathbf{k}.$$

Step 3: Compute the Flux of $\nabla \times \mathbf{F}$ Through $\mathcal{S}$

Parametrize $\mathcal{S}$ as:

$$\mathbf{r}(\theta, z) = 2 \cos \theta \mathbf{i} + 2 \sin \theta \mathbf{j} + z \mathbf{k}, \quad 0 \leq \theta \leq 2\pi, 1 \leq z \leq 6.$$

The outward-pointing normal is:

$$\mathbf{n} = 2 \cos \theta \mathbf{i} + 2 \sin \theta \mathbf{j}.$$

Evaluate $(\nabla \times \mathbf{F}) \cdot \mathbf{n}$:

$$(\nabla \times \mathbf{F}) \cdot \mathbf{n} = (0\mathbf{i} + 4z \sin(\theta)\mathbf{j} - z^2\mathbf{k}) \cdot (2 \cos \theta \mathbf{i} + 2 \sin \theta \mathbf{j} + 0\mathbf{k}).$$

Simplify:

$$(\nabla \times \mathbf{F}) \cdot \mathbf{n} = 0 + 8z \sin^2 \theta - 0 = 8z \sin^2 \theta.$$

The flux integral is:

$$\iint_{\mathcal{S}} (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS = \int_1^6 \int_0^{2\pi} 8z \sin^2 \theta \, d\theta \, dz.$$

Using $\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}$ and $\int_0^{2\pi} \sin^2 \theta \, d\theta = \pi$:

$$\iint_{\mathcal{S}} (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS = 140\pi$$

Conclusion

The line integral and the flux integral both evaluate to 140π , verifying Stokes' Theorem.

5. (14 points) SHORT QUESTIONS

- 5.1 (2 Points) Write down the statement of the Fundamental Theorem of Vector Calculus.
- 5.2 (2 Points) Explain what it means for a function $f(x)$ to be analytic at a point x_0 .
- 5.3 (2 Points) Verify if the function $f(x, y) = \sin(xy) + x - 2y$ is a solution to the partial differential equation:

$$x^2 \frac{\partial^2 f}{\partial x^2} - y^2 \frac{\partial^2 f}{\partial y^2} = 0.$$

- 5.4 (4 Points) Determine which of the following equations are first/second/third... order, linear/non-linear, exact/in-exact, separable/inseparable, and homogeneous/non-homogeneous. Indicate all classifications that apply:

1. $\sin x \cos y \, dx - \sin y \cos x \, dy = 0$
2. $0.7 \frac{d^3 \theta}{dt^3} - \theta \frac{d^2 \theta}{dt^2} + e^t \theta = 0$
3. $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + \left(\pi^2 + \frac{1}{4}\right) y - e^{-x/2} \sin(\pi x) = 0$
4. $\frac{dy}{dx} = \frac{x^2 + x}{y^2 + y}$

- 5.5 (4 Points) Let $\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$ and $\vec{G} = G_x \hat{i} + G_y \hat{j} + G_z \hat{k}$ be two vector fields. Show that:

$$\vec{\nabla} \cdot (\vec{F} \times \vec{G}) = \vec{G} \cdot (\vec{\nabla} \times \vec{F}) - \vec{F} \cdot (\vec{\nabla} \times \vec{G}).$$

Solution:

- 5.1 Let C be a smooth curve joining the point A to the point B in the plane or in space, and parametrized by $\vec{r}(t)$. Let f be a differentiable function with a continuous gradient vector $\vec{F} = \nabla f$ on a domain D containing C . Then:

$$\int_C \vec{F} \cdot d\vec{r} = f(B) - f(A).$$

- 5.2 A function $f(x)$ is said to be **analytic** at a point x_0 if it is infinitely differentiable in a neighborhood of x_0 , and its Taylor series expansion about x_0 :

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$$

converges to $f(x)$ for all x in this neighborhood. This means the function can be represented exactly by its Taylor series near x_0 .

5.3 The given function is:

$$f(x, y) = \sin(xy) + x - 2y,$$

and the PDE is:

$$x^2 \frac{\partial^2 f}{\partial x^2} - y^2 \frac{\partial^2 f}{\partial y^2} = 0.$$

Step 1: Compute Partial Derivatives

First, compute the first derivatives of $f(x, y)$:

$$\frac{\partial f}{\partial x} = y \cos(xy) + 1, \quad \frac{\partial f}{\partial y} = x \cos(xy) - 2.$$

Now, compute the second derivatives:

$$\frac{\partial^2 f}{\partial x^2} = -y^2 \sin(xy),$$

$$\frac{\partial^2 f}{\partial y^2} = -x^2 \sin(xy).$$

Step 2: Substitute into the PDE

The PDE is:

$$x^2 \frac{\partial^2 f}{\partial x^2} - y^2 \frac{\partial^2 f}{\partial y^2} = 0.$$

Substitute the second derivatives:

$$x^2(-y^2 \sin(xy)) - y^2(-x^2 \sin(xy)) = 0.$$

Simplify:

$$-x^2 y^2 \sin(xy) + y^2 x^2 \sin(xy) = 0.$$

The terms cancel, so:

$$0 = 0.$$

Thus, $f(x, y) = \sin(xy) + x - 2y$ satisfies the PDE.

5.4 Classification of Equations

(a) $\sin x \cos y \, dx - \sin y \cos x \, dy = 0$

- **Order:** First-order (only first derivatives appear implicitly).
- **Linear/Non-linear:** Non-linear (trigonometric terms are non-linear in x and y).
- **Exact/In-exact:** To check exactness:

$$M = \sin x \cos y, \quad N = -\sin y \cos x.$$

Compute:

$$\frac{\partial M}{\partial y} = -\sin x \sin y, \quad \frac{\partial N}{\partial x} = -\sin x \sin y.$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is exact.

- **Separable/Inseparable:** Inseparable.
- **Homogeneous/Non-homogeneous:** Homogeneous (degree of terms matches when scaling variables).

(b) $0.7 \frac{d^3 \theta}{dt^3} - \theta \frac{d^2 \theta}{dt^2} + e^t \theta = 0$

- **Order:** Third-order (highest derivative is third).
- **Linear/Non-linear:** Non-linear (contains the product $-\theta \frac{d^2 \theta}{dt^2}$).
- **Exact/In-exact:** Not applicable (ordinary differential equation).
- **Separable/Inseparable:** Inseparable.
- **Homogeneous/Non-homogeneous:** Homogeneous (right-hand side is 0).

(c) $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + \left(\pi^2 + \frac{1}{4}\right) y - e^{-x/2} \sin(\pi x) = 0$

- **Order:** Second-order (highest derivative is second).
- **Linear/Non-linear:** Linear (no products of y or its derivatives).
- **Exact/In-exact:** Not applicable (ordinary differential equation).
- **Separable/Inseparable:** Inseparable.
- **Homogeneous/Non-homogeneous:** Non-homogeneous (due to $-e^{-x/2} \sin(\pi x)$).

(d) $\frac{dy}{dx} = \frac{x^2 + x}{y^2 + y}$

- **Order:** First-order (only first derivatives appear).
- **Linear/Non-linear:** Non-linear (denominator contains y^2).
- **Exact/In-exact:** Not applicable (explicit form).

- **Separable/Inseparable:** Separable (can be written as $(y^2 + y)dy = (x^2 + x)dx$).
- **Homogeneous/Non-homogeneous:** Homogeneous (degree of numerator and denominator matches).

5.5 We want to prove:

$$\vec{\nabla} \cdot (\vec{F} \times \vec{G}) = \vec{G} \cdot (\vec{\nabla} \times \vec{F}) - \vec{F} \cdot (\vec{\nabla} \times \vec{G}).$$

Step 1: Expand Left-Hand Side

Using the definition of the divergence of a cross product:

$$\vec{\nabla} \cdot (\vec{F} \times \vec{G}) = \frac{\partial}{\partial x}(F_y G_z - F_z G_y) + \frac{\partial}{\partial y}(F_z G_x - F_x G_z) + \frac{\partial}{\partial z}(F_x G_y - F_y G_x).$$

Step 2: Expand Right-Hand Side

The terms on the right-hand side are:

$$\begin{aligned}\vec{G} \cdot (\vec{\nabla} \times \vec{F}) &= G_x \frac{\partial(F_z)}{\partial y} - G_x \frac{\partial(F_y)}{\partial z} + G_y \frac{\partial(F_x)}{\partial z} - G_y \frac{\partial(F_z)}{\partial x} + G_z \frac{\partial(F_y)}{\partial x} - G_z \frac{\partial(F_x)}{\partial y} \\ \vec{F} \cdot (\vec{\nabla} \times \vec{G}) &= F_x \frac{\partial(G_z)}{\partial y} - F_x \frac{\partial(G_y)}{\partial z} + F_y \frac{\partial(G_x)}{\partial z} - F_y \frac{\partial(G_z)}{\partial x} + F_z \frac{\partial(G_y)}{\partial x} - F_z \frac{\partial(G_x)}{\partial y}.\end{aligned}$$

After expanding and simplifying, the left-hand side matches the right-hand side.

Important Formulae

- Assume sufficient differentiability of the functions and that the conditions for curves and surfaces are met, then:

Del Operator $\vec{\nabla}$	$\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$
Gradient of a scalar function $\phi = \phi(x, y, z)$	$\vec{\nabla} \phi$
Divergence of a vector field $\vec{F}$	$\vec{\nabla} \cdot \vec{F}$
Curl of a vector field $\vec{F}$	$\vec{\nabla} \times \vec{F}$
Line integral of a vector field $\vec{F}$	$\int_C \vec{F} \cdot d\vec{r}$
Closed line integral	$\oint_C \vec{F} \cdot d\vec{r}$
Surface integral of vector field $\vec{F}$ over a surface S, where (u, v) are the parameters on the surface.	$\oiint_S \vec{F} \cdot \hat{n} dS = \iint_S \vec{F} \cdot \vec{N} du dv$

- For a point $P = (x, y, z)$ in Cartesian Coordinates, a representation in other coordinate systems can be

Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
$x = x$ $y = y$ $z = z$	$x = r \cos \theta$ $y = r \sin \theta$ $z = z$ where $r = \sqrt{x^2 + y^2} > 0$ $0 \leq \theta \leq 2\pi$	$x = \rho \sin \phi \cos \theta$ $y = \rho \sin \phi \sin \theta$ $z = \rho \cos \phi$ where $\rho = \sqrt{x^2 + y^2 + z^2} > 0$ $0 \leq \theta \leq 2\pi$ $0 \leq \phi \leq \pi$
$dV = dx dy dz$	$dV = r dr d\theta dz$	$dV = \rho^2 \sin \phi d\rho d\theta d\phi$

- Double Angle Identities

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}$$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}$$

- Divergence theorem for a vector field $\vec{F}$,

$$\iiint_D \vec{\nabla} \cdot \vec{F} dV = \oiint_S \vec{F} \cdot \hat{n} ds$$

Here, D is the region enclosed by the closed positively oriented surface S and V is its volume.

- Stoke's Theorem for a vector field $\vec{F}$,

$$\iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} ds = \oint_C \vec{F} \cdot d\vec{r}$$

Here the closed positively oriented curve C bounds the surface S in 3-dimensional space.

- A first order linear ODE $y' + p(x)y = r(x)$ has solution

$$y = e^{-\int p(x)dx} \left(\int e^{\int p(x)dx} r(x) dx + C \right)$$

- A first order ODE $M(x, y)dx + N(x, y)dy = 0$ is exact iff

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

- A first order non-exact ODE, $P(x, y) dx + Q(x, y) dy = 0$, can be made exact by multiplying it with an integrating function $F(x)$:

$$F(x) = e^{\int R(x) dx}, \text{ where } R(x) = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right).$$

if such a function exists.

Or, multiplying it with an integrating function $G(y)$:

$$G(y) = e^{\int R^*(y) dy}, \text{ where } R^*(y) = \frac{1}{P} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right).$$

if such a function exists.

-
- Function guesses for Method of Undetermined Coefficient,

$S(x)$	Guess for $y_p(x)$
$k = \text{Constant}$	K (Some other constant)
$ke^{\gamma x}$	$Ke^{\gamma x}$
$kx^n, (n = \text{positive integer})$	$K_n x^n + K_{n-1} x^{n-1} + \dots + K_1 x + K_0$
$k \sin \omega x$	$K \cos \omega x + M \sin \omega x$
$k \cos \omega x$	
$ke^{\alpha x} \cos \omega x$	$e^{\alpha x} (K \cos \omega x + M \sin \omega x)$
$ke^{\alpha x} \sin \omega x$	

Table 1: Short list of guesses of particular solutions for some forms of non-homogeneous part.

- The power series of $y(x)$ about $x_0 = 0$,

$$y(x) = a_0 + a_1(x) + a_2(x)^2 + a_3(x)^3 + \dots = \sum_{n=0}^{\infty} a_n x^n$$
